

ABOUT THE AUTHOR

Alan is a derivatives practitioner. This book is the route he wishes had existed when he first wanted to understand binomial option pricing without first absorbing measure theory and stochastic calculus.

He lives in the United States.

BINOMIAL OPTION PRICING

Binomial Option Pricing

From a Coin Flip to Black-Scholes

C_0 = S_0 \Phi(d_1) - Ke^{-rT} \Phi(d_2)

the Black-Scholes call price, derived in Chapter 7

INSIDE

Ch 0	Math primer (no calculus)	Ch 1	Single-period binomial
Ch 2	Multi-period tree + Greeks	Ch 3	Coin-toss probability space
Ch 4	American options & optimal stopping	Ch 5	Random walks, reflection principle
Ch 6	Bonds, caplets, callable bonds	Ch 7	Binomial → Black-Scholes (CLT)

ALAN

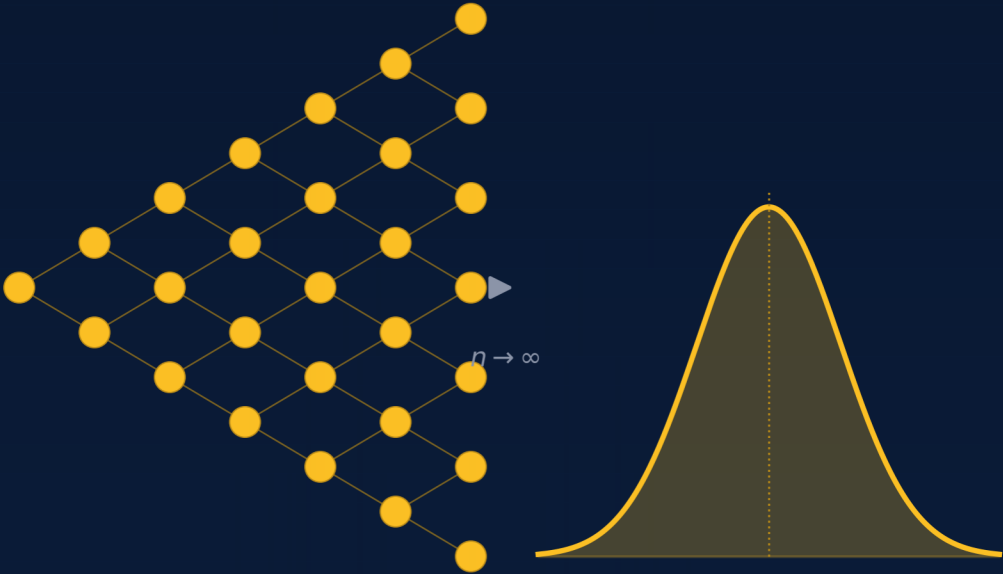
A quant-curious tour. No calculus required.

A QUANT-CURIOUS BOOK

Binomial Option Pricing

From a Coin Flip to Black-Scholes

No calculus required.



C_0 = S_0 \Phi(d_1) - Ke^{-rT} \Phi(d_2)

the Black-Scholes call price, derived in Chapter 7

ALAN

ABOUT THIS BOOK

Eight chapters take the reader from a single coin toss to the Black-Scholes formula, using only finite sums, the Central Limit Theorem stated as a fact, and a standard-normal CDF table. No derivatives, no integrals, no measure theory beyond what is genuinely needed.

Chapter 0 is a math primer with a Stirling-based proof sketch of de Moivre-Laplace. Chapters 1-6 build the binomial machine: single-period replication, multi-period trees, optimal stopping for American options, the reflection principle for path-dependents, and a short-rate tree for bonds and callable bonds. Chapter 7 — the capstone — derives Black-Scholes as the Cox-Ross-Rubinstein limit via the CLT alone.

Every concept is anchored by a worked numerical example with named numbers a reader can verify by hand.